Only in the nineteenth and twentieth centuries do we see Jews make an extraordinary contribution to humankind as a whole, through their outsized role in modern science. In addition to such well-known names as Einstein and Freud, about 20 per cent of all Nobel Prize laureates in science have been Jews, though Jews constitute less than 0.2 per cent of the world's population.[16] But it should be stressed that this has been a contribution of individual Jews rather than of Judaism as a religion or a culture. Most of the important Jewish scientists of the past 200 years acted outside the Jewish religious sphere. Indeed, Jews began to make their remarkable contribution to science only once they had abandoned the yeshivas in favour of the laboratories.

Prior to 1800, the Jewish impact on science was limited. Naturally enough, Jews played no significant role in the progress of science in China, in India or in the Mayan civilisation. In Europe and the Middle East some Jewish thinkers such as Maimonides had considerable influence on their Gentile colleagues, but the overall Jewish impact was more or less proportional to their demographic weight. During the sixteenth, seventeenth and eighteenth centuries Judaism was hardly instrumental in the outbreak of the Scientific Revolution. Except for Spinoza (who was excommunicated for his trouble by the Jewish community), you can hardly name a single Jew who was critical to the birth of modern physics, chemistry, biology or the social sciences. We don't know what Einstein's ancestors were doing in the days of Galileo and Newton, but in all likelihood they were far more interested in studying the Talmud than in studying light.

The great change occurred only in the nineteenth and twentieth centuries, when secularisation and the Jewish Enlightenment caused many Jews to adopt the world view and lifestyle of their Gentile neighbours. Jews then began to join the universities and research centres of countries such as Germany, France and the United States. Jewish scholars brought from the ghettos and shtetls important cultural legacies. The central value of education in Jewish culture was one of the main reasons for the extraordinary success of Jewish scientists. Other factors included the desire of a persecuted minority to prove its worth, and the barriers that prevented talented Jews from advancement in more anti-Semitic institutions such as the army and the state administration.

Yet while Jewish scientists brought with them from the yeshivas strong discipline and a deep faith in the value of knowledge, they did not bring any helpful baggage of concrete ideas and insights. Einstein was Jewish, but the theory of relativity wasn't 'Jewish physics'. What does faith in the sacredness of the Torah have to do with the insight that energy equals mass multiplied by the speed of light squared? For the sake of comparison, Darwin was a Christian and even began his studies at Cambridge intending to become an Anglican priest. Does it imply that the theory of evolution is a Christian theory? It would be ridiculous to list the theory of relativity as a Jewish